

Cyber Security CTF Machine Development Report

Title:

Cyber security CTF Machine Development Assignment

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Course: Cyber Security

Operating System :kali Linux

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1. Objective

The objective of this assignment was to design and deploy a vulnerable Capture The Flag (CTF) machine on a Linux environment. The machine simulates a real-world penetration testing target where participants perform reconnaissance, enumeration, web exploitation, SSH access, privilege escalation, and steganography to capture flags. The machine is intentionally configured with hidden clues that require logical analysis instead of direct disclosure of credentials.

2. Machine Specifications

Specification Details

Virtualization Software: VMware Workstation

Development Environment: Kali Linux

Target Machine: Ubuntu 14.04.6 LTS

Web Server: Apache2

SSH Service: OpenSSH

Hostname: madtha

Firewall: UFW

Browser Tested: Firefox

3. Operating System Used

The operating system used for developing the CTF machine is **Kali Linux**, which provides various tools required for penetration testing and security analysis. The target machine was configured using **Ubuntu 14.04.6 LTS**, which served as the vulnerable system for the CTF challenge.

The combination of Kali Linux and Ubuntu created a realistic environment for performing reconnaissance, enumeration, exploitation, privilege escalation, and steganography-based challenges.

4. Network Configuration

The machine was configured using NAT networking allowing communication between the attacker and target systems.

Accessible ports:

Port 22 (SSH)

Port 80 (HTTP)

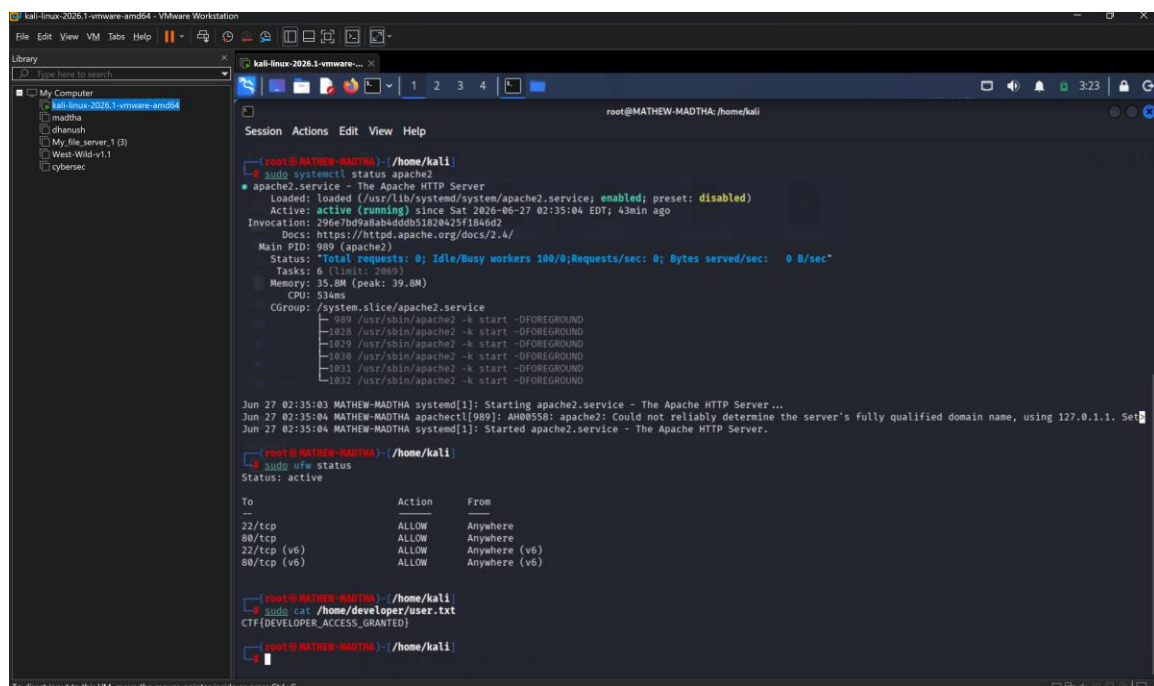
All unnecessary ports were blocked using the firewall.

5. User Accounts Created

Three separate Linux user accounts were created.

- mathew
- mokshith
- neha

Each user has an individual home directory and login shell. Credentials were hidden inside the challenge environment.



```
kali-linux-2026.1-vmware-amd64 - VMware Workstation
File Edit View VM Tabs Help
Library
My Computer
kali-linux-2026.1-vmware-amd64
maditha
dhanush
My file_server_1 (3)
West-Wild-v1.1
cybersec

root@MATHIEW-MADTHA:/home/kali
Session Actions Edit View Help

root@MATHIEW-MADTHA:~/home/kali
# sudo systemctl status apache2
● apache2.service - The Apache HTTP Server
   Loaded: loaded (/usr/lib/systemd/system/apache2.service; enabled; preset: disabled)
   Active: active (running) since Sat 2026-06-27 02:35:04 EDT; 43min ago
     Invocation: 296e7bd9a8ab4ddb51820425f1846d2
       Docs: https://httpd.apache.org/docs/2.4/
    Main PID: 989 (apache2)
      Status: "Total requests: 0; Idle/Busy workers 100/0; Requests/sec: 0; Bytes served/sec: 0 B/sec"
        Tasks: 6 (limit: 2050)
      Memory: 35.8M (peak: 39.8M)
         CPU: 534ms
    CGroup: /system.slice/apache2.service
            └─ 989 /usr/sbin/apache2 -k start -DFOREGROUND
               1028 /usr/sbin/apache2 -k start -DFOREGROUND
               1029 /usr/sbin/apache2 -k start -DFOREGROUND
               1030 /usr/sbin/apache2 -k start -DFOREGROUND
               1031 /usr/sbin/apache2 -k start -DFOREGROUND
               1032 /usr/sbin/apache2 -k start -DFOREGROUND

Jun 27 02:35:03 MATHIEW-MADTHA systemd[1]: Starting apache2.service - The Apache HTTP Server...
Jun 27 02:35:04 MATHIEW-MADTHA apache2[989]: AH00558: apache2: Could not reliably determine the server's fully qualified domain name, using 127.0.1.1. Set
Jun 27 02:35:04 MATHIEW-MADTHA systemd[1]: Started apache2.service - The Apache HTTP Server.

root@MATHIEW-MADTHA:~/home/kali
# sudo ufw status
Status: active

To Action From
--
22/tcp ALLOW Anywhere
80/tcp ALLOW Anywhere
22/tcp (v6) ALLOW Anywhere (v6)
80/tcp (v6) ALLOW Anywhere (v6)

root@MATHIEW-MADTHA:~/home/kali
# sudo cat /home/developer/user.txt
CTF{DEVELOPER_ACCESS_GRANTED}

root@MATHIEW-MADTHA:~/home/kali
```

6. Services Configured

The following services were configured.

Apache Web Server

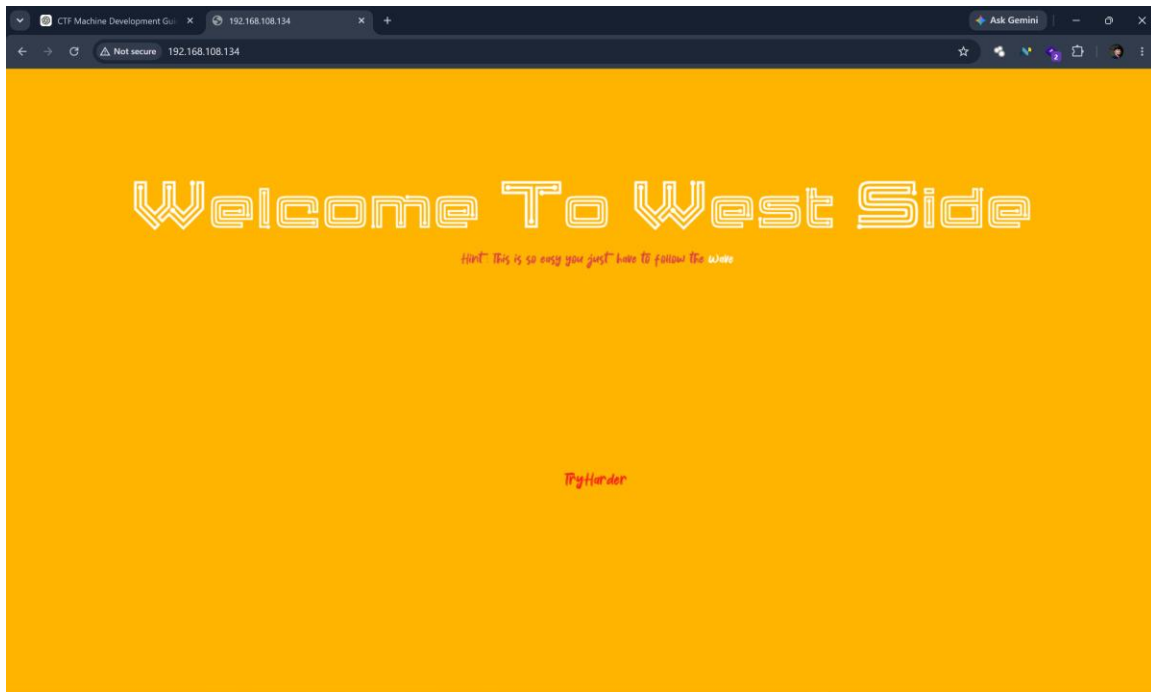
- Runs on Port 80
- Hosts portfolio website
- Contains hidden clues

SSH

- Runs on Port 22
- Used after discovering credentials

Firewall

- Configured using UFW allowing SSH and HTTP



7. Website Design Overview

A personal portfolio website was developed containing:

- Professional profile
- Skills
- Projects
- Contact information

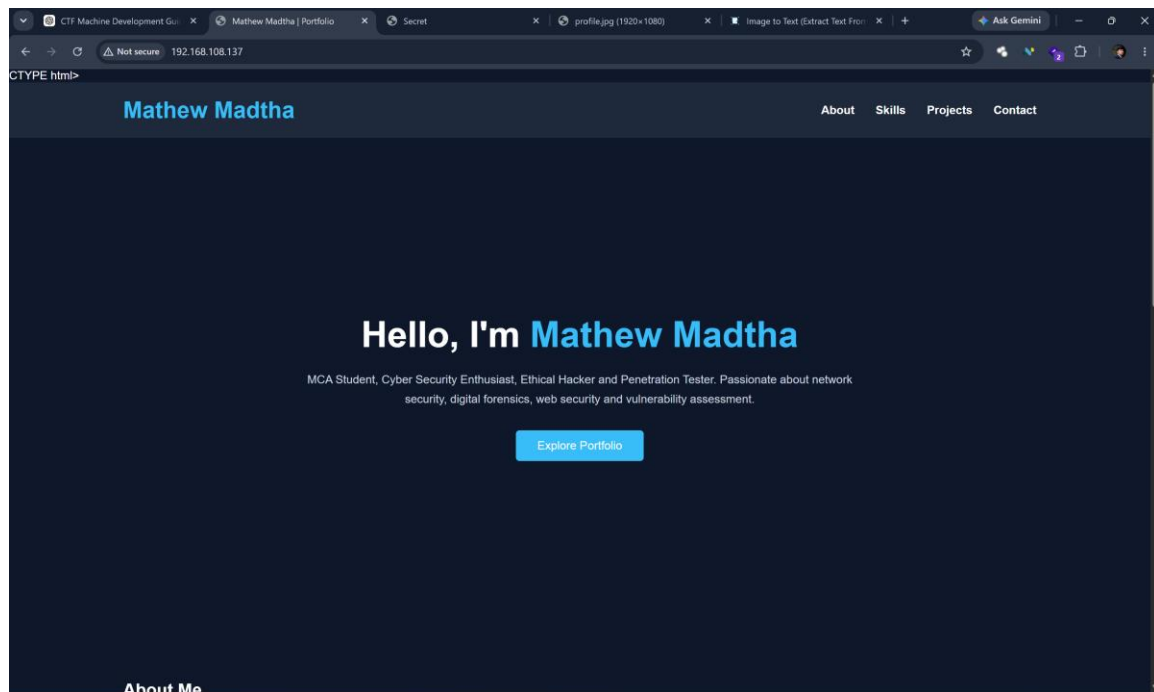
The website contains intentionally hidden information for participants to discover.

Website technologies:

HTML

CSS

JavaScript

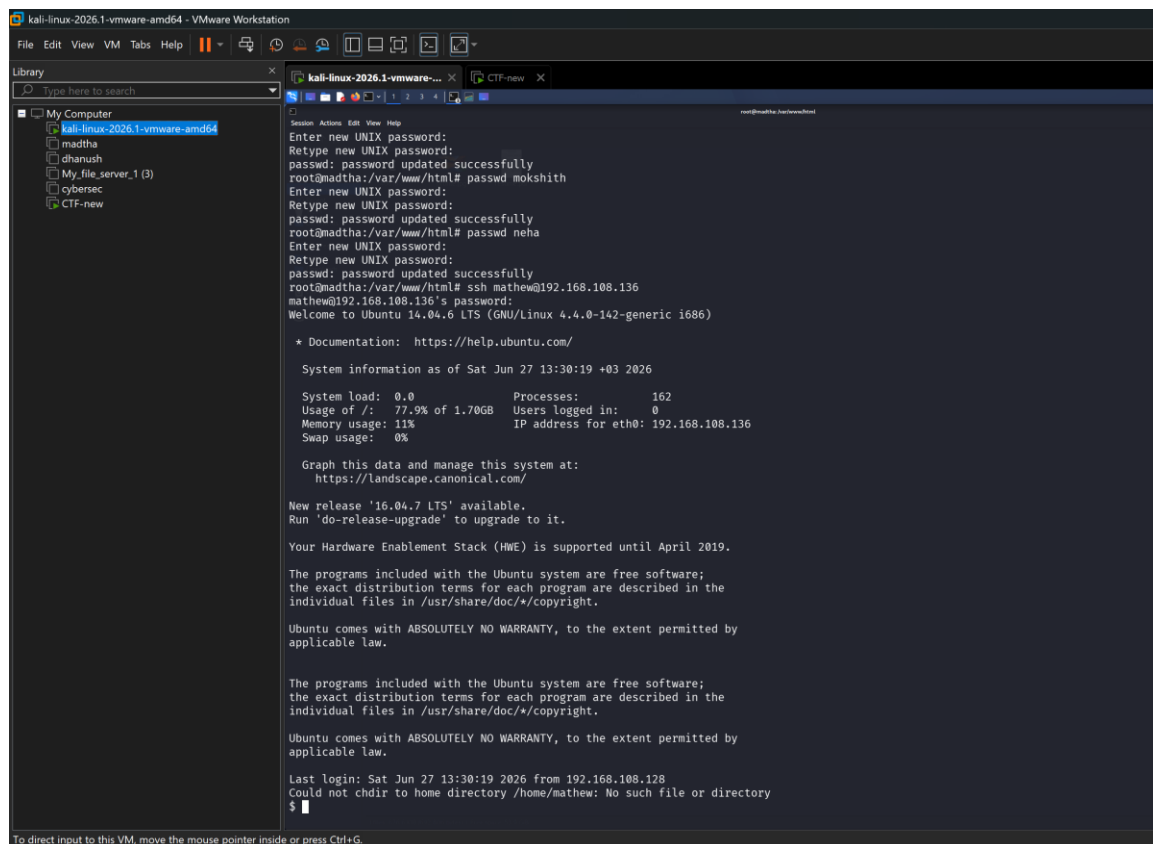


8. Hidden Credentials Methodology

The machine requires enumeration.

Hidden information was distributed using:

- HTML comments
- JavaScript source files
- CSS comments
- robots.txt
- Hidden directories
- Encoded strings (Base64)
- Source code inspection



The screenshot shows a Kali Linux virtual machine running on VMware Workstation. The terminal window displays the following output:

```
kali-linux-2026.1-vmware-amd64 - VMware Workstation
File Edit View VM Tabs Help
Library
Type here to search
My Computer
kali-linux-2026.1-vmware-amd64
madtha
dhanush
My_file_server_1 (3)
cybersec
CTF-new
root@madtha:~#
Enter new UNIX password:
Retype new UNIX password:
passwd: password updated successfully
root@madtha:/var/www/html# passwd mokshith
Enter new UNIX password:
Retype new UNIX password:
passwd: password updated successfully
root@madtha:/var/www/html# passwd neha
Enter new UNIX password:
Retype new UNIX password:
passwd: password updated successfully
root@madtha:/var/www/html# ssh mathew@192.168.108.136
mathew@192.168.108.136's password:
Welcome to Ubuntu 14.04.6 LTS (GNU/Linux 4.4.0-142-generic i686)

 * Documentation:  https://help.ubuntu.com/

System information as of Sat Jun 27 13:30:19 +03 2026

System load:  0.0          Processes:    162
Usage of /:   77.9% of 1.70GB Users logged in:  0
Memory usage: 11%         IP address for eth0: 192.168.108.136
Swap usage:   0%

Graph this data and manage this system at:
https://landscape.canonical.com/

New release '16.04.7 LTS' available.
Run 'do-release-upgrade' to upgrade to it.

Your Hardware Enablement Stack (HWE) is supported until April 2019.

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

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applicable law.

Last login: Sat Jun 27 13:30:19 2026 from 192.168.108.128
Could not chdir to home directory /home/mathew: No such file or directory
$
```

To direct input to this VM, move the mouse pointer inside or press Ctrl+G.

9. Steganography Implementation

A file named profile.jpg was used for steganography. A professionally formatted resume.pdf was embedded into the image using steghide.

Example extraction:

steghide extract -sf profile.jpg

```
steghide: the passphrases do not match.
root@madtha:/home/mathew# steghide embed -cf profile.jpg -ef resume.pdf
Enter passphrase:
Re-Enter passphrase:
embedding "resume.pdf" in "profile.jpg"... done
root@madtha:/home/mathew# steghide info profile.jpg
"profile.jpg":
  format: jpeg
  capacity: 35.1 KB
Try to get information about embedded data ? (y/n) y
Enter passphrase:
embedded file "resume.pdf":
  size: 24.7 KB
  encrypted: rijndael-128, cbc
  compressed: yes
root@madtha:/home/mathew# echo "My professional profile is hidden inside my photograph." > /var/www/html/hint.txt
root@madtha:/home/mathew# cat /var/www/html/hint.txt
My professional profile is hidden inside my photograph.
root@madtha:/home/mathew# cp profile.jpg /var/www/html/
root@madtha:/home/mathew# ls -l /var/www/html/profile.jpg
-rw-r--r-- 1 root root 663653 Jun 27 14:57 /var/www/html/profile.jpg
root@madtha:/home/mathew# nano /var/www/html/secret.html
root@madtha:/home/mathew# cp /home/mathew/profile.jpg /var/www/html/
root@madtha:/home/mathew# echo "Disallow: /secret.html" > /var/www/html/robots.txt
root@madtha:/home/mathew# cat /var/www/html/robots.txt
Disallow: /secret.html
root@madtha:/home/mathew# ls -l /var/www/html/profile.jpg
-rw-r--r-- 1 root root 663653 Jun 27 14:59 /var/www/html/profile.jpg
root@madtha:/home/mathew#
```


10. Flag Locations and Hostname

Two flags were created.

User Flag:

/home/wavex/wave/FLAG1.txt

Root Flag:

/root/FLAG2.txt

Hostname:

madtha

11. Screenshots of Configuration

Include the following screenshots:

- Hostname configuration
- User accounts
- Apache service status
- SSH service status
- Firewall configuration
- Website homepage
- robots.txt
- Hidden page
- Steganography information
- User flag
- Root flag

12. Hostname Configuration

The system hostname was configured as madtha to uniquely identify the target machine within the virtual network environment. The hostname configuration helps distinguish the CTF machine from other systems during reconnaissance and penetration testing activities.

The hostname was verified using Linux system commands and was used throughout the configuration and testing phases of the project. Assigning a unique hostname provides better system identification and management in a controlled cybersecurity environment.

Configured Hostname: madtha

13. Testing Procedure

Step 1: Port scanning using nmap

Step 2: Visit website

Step 3: Inspect source code

Step 4: Check robots.txt

Step 5: Enumerate hidden files

Step 6: Decode hidden strings

Step 7: Recover credentials

Step 8: SSH Login

Step 9: Capture user flag

Step 10: Privilege escalation

Step 11: Capture root flag

Step 12: Extract hidden resume

14. Vulnerability Walkthrough

1. Perform network scan to identify open services.
2. Access the hosted portfolio website.
3. Inspect HTML, CSS, JavaScript, and robots.txt for hidden clues.
4. Decode hints to discover credentials.
5. Authenticate via SSH.
6. Retrieve the user flag.
7. Escalate privileges.
8. Retrieve the root flag.
9. Extract the embedded resume using steganography.

15. Conclusion

This project demonstrates the design and deployment of a realistic Capture The Flag machine using Kali Linux. The machine integrates web enumeration, credential discovery, SSH authentication, privilege escalation, firewall configuration, and steganography into a single practical exercise.

Appendix

Software Used:

Kali Linux

Ubuntu 14.04.6 LTS

Apache2

OpenSSH

UFW Firewall

Steghide

VMware Workstation

Firefox Browser